

N109AV (S/N: W03021891423) — Modem Hardware

Failure Diagnostic Report

Immfly Ground Operations / IT Session: September 15, 2026

Field	Value
Machine	N109AV (S/N: W03021891423)
Affected component	Sierra Wireless EM7565 Cellular Modem (USB Bus 1-4.1 disconnect / dead state)
Initial symptom	No cellular connectivity; modem_fix.py detected outdated firmware (SWI9X50C_01.08.04.00) in MBIM mode, skipped upgrade, and subsequently lost all hardware communication.
USB identification	ABSENT — Completely missing from USB bus topology (no 1199:9091 nor 1199:9090 detected by kernel).
Root cause	Total modem hardware/electrical failure or permanent PCIe/M.2 bus power cutoff. Device does not respond to PCI bus rescan, USB driver unbind/bind, nor a full cold machine reboot. rfkill reports no blocks.
Comparison	Distinct from recovery mode (N136AV / N060AV) and authentication failures (N029AV). The module in N109AV has completely ceased electrical enumeration on USB port 1-4.1.
Recommended action	Physical modem replacement required. Software-level recovery exhausted.

NARRATIVE SUMMARY

N109AV exhibited total failure of cellular connectivity. Diagnostic execution via modem_fix.py initially reported an outdated firmware version (SWI9X50C_01.08.04.00) operating in MBIM mode, but skipped the remediation process. Immediately following this cycle, ModemManager reported No modems were found.

Low-level kernel inspection with lsusb and dmesg revealed that the Sierra Wireless module (Vendor ID 1199) had vanished entirely from internal USB hub 1-4.1. Nearby peripheral devices on the same physical controller (FTDI serial adapters, Teensyduino on port 1-4.3) remained fully operational, localizing the fault specifically to the modem port/socket.

Subsequent recovery attempts including kernel PCI bus rescan (echo 1 > /sys/bus/pci/rescan) and forced USB driver unbind/bind on bus 1-4 failed to revive the module. A full host system reboot was performed to power-cycle the internal bus. Post-reboot, the modem still failed to enumerate at the electrical layer, and rfkill list showed no active software or hardware RF kill switches. The modem module is completely unresponsive.

N109AV (S/N: W03021891423) — Troubleshooting Steps & Command Appendix

TROUBLESHOOTING STEPS PERFORMED

- Automated diagnostic & initial state check:**
modem_fix.py reported old firmware (SWI9X50C_01.08.04.00) and MBIM mode, but skipped remediation and failed to establish PPP session.
- ModemManager daemon interrogation:**
mmcli -L returned "No modems were found". Restarting ModemManager.service yielded identical results.
- USB bus & topology audit:**
lsusb | grep -i 1199 returned empty output. dmesg verified all other FTDI and Teensy serial bridges were operational on hub 1-4, but port 1-4.1 had zero attached devices.
- Software bus cycle & PCI rescan:**
Executed echo 1 > /sys/bus/pci/rescan and unbind/bind cycle on sysfs node /sys/bus/usb/drivers/usb/1-4. The Sierra device failed to negotiate or signal handshake.
- Full machine cold reboot:**
Host was rebooted via reboot. Post-boot verification confirmed the modem remained completely absent from lsusb and dmesg.
- RF kill switch verification:**
rfkill list inspected to rule out PCIe pin-disable; only Bluetooth and WLAN interfaces were present (all unblocked).

CONCLUSION AND RECOMMENDATION

N109AV's Sierra Wireless EM7565 modem is electrically dead or permanently severed from the internal USB/M.2 bus. The issue persists across hardware re-enumeration and full system power cycles, confirming an unrecoverable hardware failure.

Recommended action: Issue a maintenance work order for **physical replacement of the Sierra Wireless modem module** in aircraft N109AV.

APPENDIX: KEY COMMAND OUTPUT

```
[AVA:N109AV] mmcli -L & lsusb check:
# mmcli -L
No modems were found

# lsusb | grep -i 1199
(no output)

# lsusb
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 006: ID 16c0:0483 Van Ooijen Technische Informatica Teensyduino Serial
Bus 001 Device 005: ID 0403:6015 Future Technology Devices International, Ltd Bridge(I2C/SPI/UART/FIFO)
Bus 001 Device 004: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 001 Device 003: ID 8087:0025 Intel Corp.
Bus 001 Device 002: ID 0403:6010 Future Technology Devices International, Ltd FT2232C/D/H Dual UART/FIFO IC
```

```
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

[AVA:N109AV] Kernel topology & rfkill state post-reboot:

```
# dmesg | grep -i -E "1199|sierra|1-4.1"
[  5.317116] hub 1-4:1.0: USB hub found
[  5.321205] hub 1-4:1.0: 4 ports detected
```

```
# rfkill list
0: hci0: Bluetooth
    Soft blocked: no
    Hard blocked: no
1: phy0: Wireless LAN
    Soft blocked: no
    Hard blocked: no
```